



Influence of fermentation conditions on the sensorial quality of coffee inoculated with yeast

Marcela Caroline Batista da Mota^a, Nádia Nara Batista^a, Mariane Helena Sances Rabelo^b,
Diego Egidio Ribeiro^c, Flávio Meira Borém^b, Rosane Freitas Schwan^{a,*}

^a Department of Biology, Federal University of Lavras, Lavras, MG, Brazil

^b Department of Engineering, Federal University of Lavras, Lavras, MG, Brazil

^c Syngenta Crop Protection – Nucoffee, Varginha, MG, Brazil



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ABSTRACT

Fermentation is one of the post-harvest steps that influence coffee quality. This work evaluated the effect of *Saccharomyces cerevisiae* (CCMA 0543) and *Torulaspora delbrueckii* (CCMA 0684) inoculation on the quality of natural and pulped natural processed coffee in different producing regions. Yeast populations were assessed by the real-time polymerase chain reaction. Volatile and nonvolatile compounds were evaluated by gas chromatography-mass spectrometry and high-performance liquid chromatography, respectively. *S. cerevisiae* was predominant during spontaneous (average of 4 log cells/g) and inoculated (average of 7 log cells/g) fermentations in both processes. *T. delbrueckii* showed a similar population (3.79 log cells/g) in all assays. Glucose and fructose were the most detected sugars in coffee beans. Succinic acid was found at the end of the fermentative process. The lactic acid concentration was inversely proportional to ethanol concentration. Pyrrole and furan, which are volatile groups, allowed to differentiate the coffee processing methods. Yeast inoculation modified the sensorial profile and increased the coffee beverage scores by up to 5 points. *S. cerevisiae* inoculation was most suitable for pulped natural coffee, and *T. delbrueckii* inoculation showed the best performance in natural coffee.

1. Introduction

Coffee is an important commercial product that has a strong economic influence in several underdeveloped countries (CONAB, 2019). Brazil is a major global coffee producer and exporter, and in 2018, it produced 61.7 million bags of coffee, followed by Vietnam, Colombia, Indonesia, Ethiopia, and India (CONAB, 2019; OIC, 2018).

Coffee production with different sensory profiles has attracted the attention of many consumers. The sensorial perceptions are influenced by genetic traits, geographical factors, agricultural practices, and post-harvest techniques (Borrelli, Mataix, & Carrasco-Gallego, 2015).

Post-harvesting processing includes fermentation, drying, and storage. The coffee fermentation process can be performed in three different ways: natural method, wet method, and pulped natural (Farah, 2019). (i) In the natural method, newly harvested whole fruits are transferred to a drying platform where the fruits are fermented and dried. (ii) In the wet method, the exocarp and part of the mesocarp are mechanically removed (Brando & Brando, 2014). Parchment coffee is transferred to water tanks, where the remaining mucilage is degraded and solubilized during the fermentative process (6–72 h) (Brando &

Brando, 2014). The coffee beans are then removed from the tanks and sun-dried. (iii) In the pulped natural, the coffee peel and part of or all the mucilage are removed mechanically, and the coffee is transferred to a drying platform where the beans are fermented and dried up to 11% of moisture content (Brando & Brando, 2014; Vilela, de Pereira, Silva, Batista, & Schwan, 2010).

The different processing methods and environmental conditions modify the microbial behavior, volatile precursors, and organic acid production of green coffee beans, contributing to modifications in the volatile and aroma profiles of the roasted coffee (Bressani, Martinez, Evangelista, Dias, & Schwan, 2018; Evangelista et al., 2014; Lee, Cheong, Curran, Yu, & Liu, 2015; Martinez, Bressani, da Miguel, Dias, & Schwan, 2017; Ribeiro et al., 2017a; Ribeiro et al., 2017b). Yeast has an essential role in the fermentative process because of its biocontrol of filamentous fungi growth, further producing different enzymes and volatile aromatic compounds (Souza et al., 2017).

The inoculation of a starter culture was shown to improve coffee flavor and aroma, reduce processing time and drying time, and increase the economic value of the product (Evangelista et al., 2014; Silva et al., 2013). *Saccharomyces cerevisiae* (CCMA 0543) and *Torulaspora*

* Corresponding author at: Federal University of Lavras, Department of Biology, Campus Universitário, 3037, 37200-900 Lavras, MG, Brazil.

E-mail addresses: flavioborem@ufla.br (F.M. Borém), rschwan@ufla.br (R.F. Schwan).

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delbrueckii (CCMA 0684), isolated from natural coffee fermentation, were previously selected as a starter culture for coffee fermentation because of their high pectinolytic activity, excellent fermentative capacity, and production of desirable metabolic in the final product such as acids, alcohols, pyridines, aldehyde, and furans (Silva et al., 2013; Evangelista et al., 2014a; Evangelista et al., 2014b). The inoculation of these yeasts has improved the quality of the beverage, and some attributes have been found through it, such as caramel, chocolate, herbaceous materials, yellow fruits, and almonds.

This work aimed to evaluate the dynamic behavior of *S. cerevisiae* (CCMA 0543) and *T. delbrueckii* (CCMA 0684) inoculated in coffee from different geographical origins using natural and pulped natural processes.

2. Materials and methods

2.1. Fermentation experiments and sampling

For the fermentation experiments, the five largest Arabica coffee-producing regions with different climate profiles were selected: Carmo de Minas, Três Pontas, Araxá, Monte Carmelo, and Lajinha, Minas Gerais State, Brazil (Table 1). Coffee fruits from Bourbon Amarelo, Catucaí, and Rubi were harvested from May to August 2018.

The coffee was harvested manually and processed using natural and pulped natural methods. Approximately 40 L of coffee was fermented in high-density polyethylene bioreactors, cylindrical with capacity for 50L (Picillo, São Paulo, Brazil). Fermentation was performed both with and without inoculation (control) of the yeast starter culture (*S. cerevisiae* [CCMA 0543] and *T. delbrueckii* [CCMA 0684]) at the beginning of the process. The strains were grown separately in YPD broth (10 g/L yeast extract [Merck], 20 g/L peptide [Himedia], and 20 g/L dextrose [Merck]) at 30 °C at 150 rpm and replicated every 24 h (Evangelista et al., 2014b). The cells were recovered by centrifugation (7,000 rpm, 10 min) and resuspended in 500 mL peptone water (1 g/L of Peptona [Himedia]). This solution was spread over the coffee beans to a concentration of approximately 10^7 cells/g of coffee. Samples of approximately 100 g each were withdrawn at 18 °C, 22 °C, and 30 °C of the process (approximately 0, 30, and 70 h). The temperature matched the beginning, middle, and end of the fermentation, respectively. The samples were placed in sterile plastic bags, transferred to the laboratory, and stored at – 20 °C for further chemical and microbiological analysis. All fermentations were performed in triplicate.

2.2. DNA extraction and real-time qualitative polymerase chain reaction (qPCR)

For DNA extraction, 3 g of coffee was weighed and diluted with 5 mL Milli-Q water and vortexed for 5 min. The supernatant was collected in a Falcon tube and centrifuged (9,000 rpm, 10 min, 4 °C). The

supernatant was separated from the precipitate. Total DNA was extracted from 50 mg of pellets as previously described (Cocolin, Bisson, and Mills (2000)). The DNA was stored at – 20 °C for further use.

Specific primers for *S. cerevisiae* (SC-5fw5'-AGGAGTGC GGTTCTTTCTAAAG-3' and SC-3bw5'-TGAAATGCGAGATTC CCCCCA-3') and *T. delbrueckii* (L25'-CAAAGTCATCCAAGCCAGC and R25'-TTCT CAAACAATCATGTTTGGTAG-3') were used in this study (Díaz, Molina, Nähring, & Fischer, 2013). Real-time qPCR was performed using the Rotor-Gene Q system (Qiagen, Hombrechtikon, ZH, Switzerland). Each reaction comprised 10 µL of Rotor-Gene QuantiNova Sybr Green master mix (Qiagen, Stockach Konstanz, Germany), 2 µL (12.5 µM) of each primer (Invitrogen, Sao Paulo, SP, Brazil), and 1 µL of DNA extracted from the coffee beans for a total volume of 20 µL. The mixture was heated at 95 °C for 2 min, followed by 40 cycles of denaturation at 95 °C for 5 s and annealing/extension at 60 °C for 10 s. The cycling temperature was increased by 1 °C every 5 s from 50 °C to 99 °C to obtain the melting curve. All analyses were performed in triplicate. For the standard curves, all yeast species were cultivated in YPD agar at 28 °C for 24 h. The cells were counted using a Neubauer chamber. DNA was extracted using the methodology described by Cocolin et al. (2000) and serially diluted (1:10) from 10^8 to 10^3 cells/mL down to 10 cells/mL (Batista, Ramos, Ribeiro, Pinheiro, & Schwan, 2015). Each point on the calibration curve was measured in triplicate.

2.3. Analysis of carbohydrates, alcohols, and organic acids

Carbohydrates, alcohols, and organic acids from coffee beans were extracted (Ribeiro et al., 2017a) and analyzed using a liquid chromatography system (Shimadzu, model LC-10Ai, Shimadzu Corp., Japan) equipped with an ultraviolet (UV)-visible (SPD 10Ai) and refractive index (RID-10Ai) detection system. The Shim-pack SCR-101H ion exclusion column (7.9 mm × 30 cm) was used to detect acids at 50 °C at a flow rate of 0.6 mL/min of perchloric acid (100 mM). Carbohydrate and alcohol detection were performed using the Shim-pack SCR-101C column at 80 °C with a 0.6 mL/min flow of deionized Milli-Q water. The acids were detected by UV absorbance (210 nm), and RID detected the carbohydrates and alcohols. Compound identification was based on the retention time of the standards injected using the same conditions. The sample concentration was determined using external calibration. Calibration curves were obtained by injecting different concentrations of the standards, and the areas obtained were used to plot a linear curve, the equation of which was used to estimate the concentration of the compounds in the samples (Evangelista et al., 2014). All samples were analyzed in triplicate.

2.4. Gas chromatography-mass spectrometry (GC-MS) for volatile compound analysis

Volatile compounds were extracted from the roast coffee (3 g) using the solid-phase microextraction technique in the headspace (SPME-HS). Fiber (divinylbenzene/carboxy/polydimethylsiloxane HS-SPME 50/30 µm; Supelco Co., Bellefonte, PA, USA) was used to extract the volatile compounds. The fiber was equilibration for 15 min at 60 °C and then exposed to the ground roast coffee for 30 min at the same temperature (Visintin et al., 2017).

Volatile compounds were analyzed by gas chromatograph-mass spectrometry (GCMS-QP 2010, Shimadzu) equipped with an Agilent HP-FFAP column (30 m × 0.25 mm × 0.25 µm). The oven temperature was set to 50 °C for 5 min, increased to 190 °C at a rate of 3 °C min⁻¹, and maintained at 190 °C for 10 min. The carrier gas was He at 1.2 mL/min. The splitless injection mode was at 240 °C, and the mass detector was a quadrupole with an electronic impact ionization system at 70 eV and 260 °C. Compounds were identified using the GC-MS solution (version 2.6) software. The mass spectra for each peak were evaluated in the NIST11 database. All samples were analyzed in triplicate.

Table 1
Geographic characteristics of coffee-producing regions.

	Southern of Minas Gerais		Cerrado		Zona da Mata
	Carmo de Minas	Três Pontas	Araxá	Monte Carmelo	Lajinha
Coffee variety	Bourbon amarelo	Catucaí amarelo	Bourbon amarelo	Rubi	Catucaí amarelo
Location	1	2	3	4	5
Latitude	22° 6' S	21° 22' S	19° 35' S	18° 38' S	20° 09' S
Longitude	45° 8' W	45° 30' W	46° 56' W	47° 27' W	41° 37' W
Altitude	1.161 m	885 m	997 m	963.35 m	470 m
Average annual temperature	19.1 °C	20.0 °C	20.2 °C	21.2 °C	22.0 °C
Rainfall	1.568 mm	1.440 mm	1.626 mm	1.444 mm	1.194 mm

Sample	Flavor		Taste		Acidity	Body	Finishing	Sweetness	Acidity	Body	Finishing	Defects	Final Note
	Citrus fruits	Winey	Tropical fruits	Milk chocolate	Citrus	Creamy	Short					Riado	
	Tropical fruits	Coffee pulp	Dehydrated fruits	Bitter chocolate		Phosphoric						Viscous	
	Dry fruits		Yellow fruits	Dairy Notes									
	Yellow fruits	Alcoholic	Red fruits	Chestnut and Almonds	Malic	OilY	Average					River	
	Red holes		Lemon Cloves	Woody									
	Floral	Vinegar	Sicilian Lemon	Spices	Acetic	Silky						River zone	
	Honey		Orange	Winey									
	Caramel	Black tea	Coffee flower	Coffee pulp			Unwanted fermented						
	Brown sugar		Orange flower	Alcoholic									
	Caramelized Sugar	Tobacco	Jasmine	Vinegar	Lactic	Liqueur	Long						
	Milk chocolate		Honey	Black tea									
	Bitter chocolate	Earthy	Sucrose	Tobacco	Tartaric	Others						Absent	
	Dairy Notes		Brown sugar	Earthy									
	Chestnuts and almonds	Peanut	Caramelized Sugar	Peanut			Uniform (N°)						
	Woody		Sugar cane	Herbace									
Spices	Herbace	Caramel											
Comments:													

Fig. 1. Sensory assessment sheet.

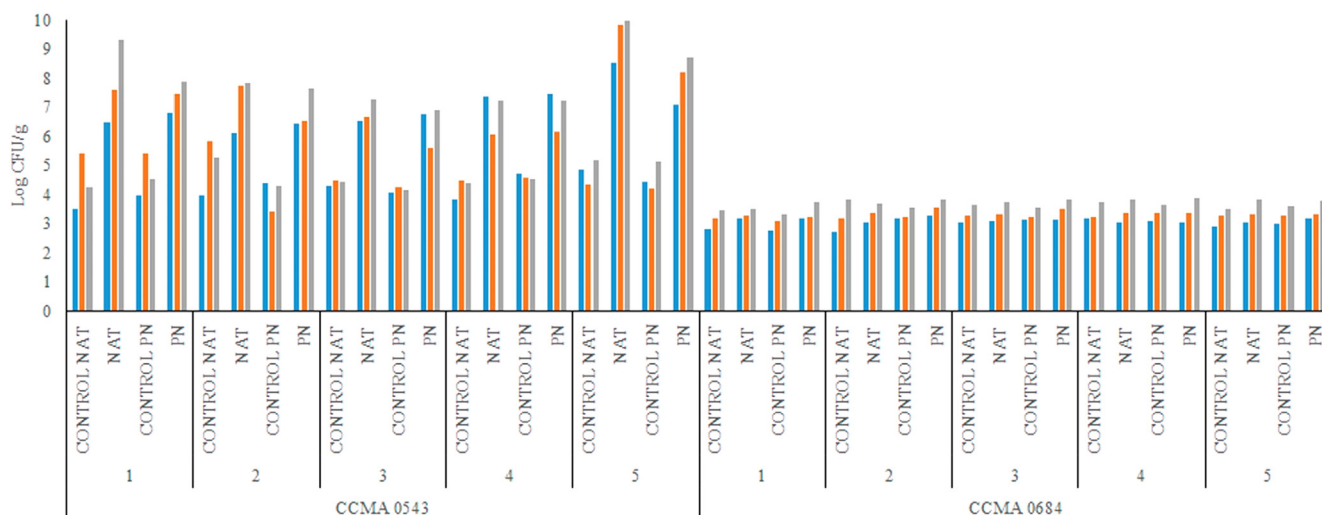


Fig. 2. Dynamic population of *S. cerevisiae* and *T. delbrueckii* in beginning, middle and end spontaneous (control) and inoculated fermentation of natural coffee (NAT) and pulped natural coffee (PN) in region 1 (a), region 2 (b), region 3 (c), region 4 (d), and region 5 (e) measured by qPCR.

2.5. Sensory analysis

The coffee samples were prepared according to the Specialty Coffee Association (SCA, 2013). The coffee samples (100 g) were roasted in a Probat Leogap roaster (model TP2) with a capacity of 150 g and then grounded in an electric mill (Mahlkönig model, Guatemala). Six expert coffee tasters performed the sensory analysis with a Q-Grader Coffee Certificate performed in five cups of each sample. The Q-Graders were asked to select the sensory attributes that they considered appropriate to describe the coffee (Fig. 1). An unstructured scale was used to evaluate sweetness intensity, acidity, body, and finish. All analyses were performed in triplicate.

2.6. Statistical analysis

The fermentation process was conducted in a randomized block design (DBC), with each block representing a region of Minas Gerais State. The treatments were evaluated in a $2 \times 3 \times 5$ factorial

arrangement with two treatments (natural and pulped natural), three treatments (*S. cerevisiae* [CCMA0543], *T. delbrueckii* [CCMA 0684], and control), and five regions (Carmo de Minas, Três Pontas, Araxá, Monte Carmelo, and Lajinha), and they were performed in triplicate. Heatmap and principal component analysis (PCA) were performed using the XLSTAT 2019 2.1 software for the volatile group and sensory analysis, respectively.

3. Results and discussion

3.1. Fermentative process

Agricultural and post-harvest practices are correlated with the quality of the coffee beverage. Controlling the fermentation process using starter cultures and monitoring the temperature ensures well-fermented coffee. During fermentation, the microbial growth favored increasing fermentation temperature, indicating that the fermentation process has been adequately conducted (Jackels & Jackels, 2005;

Peñuela-Martínez, Carlos, Oliveros-Tascón, Juan, & Sanz-Urbe, 2010). Previous work showed that the stabilization or decrease in the temperature during the coffee fermentation process is a parameter that defines the end of the fermentation process (Martins et al., 2020; Peñuela-Martínez et al., 2010).

Starter cultures, *S. cerevisiae* (CCMA 0543), and *T. delbrueckii* (CCMA 0684) strains were monitored during natural and pulped coffee fermentation in the two assays (control and inoculated) by qPCR (Fig. 2).

S. cerevisiae was predominant in both processing methods. In control, *S. cerevisiae* reached 5.69 log CFU/g and *T. delbrueckii* 4.85 log CFU/g. *S. cerevisiae* can adapt quickly to different environmental conditions (Legras et al., 2018). Its ability to ferment coffee is associated with the presence of enzymes, such as pectin lyase and polygalacturonase, which contribute to the increased concentration of carbon sources (Silva, Batista, Abreu, Dias, & Schwan, 2008; Silva et al., 2013; Ribeiro et al., 2018).

Different environmental conditions influenced the increase in the yeast population. *S. cerevisiae* (CCMA 0543) inoculated in coffee reached a population of 7.0 log CFU/g at the end of fermentation, except in regions 1 (coffee natural) and 5 (pulped natural), which had a population of 9.0 log CFU/g at the end of fermentation.

Fermentation with and without inoculation of *T. delbrueckii* showed a similar population (4.85 and 3.79 log cells/g) at the end of the fermentation process in all regions. The low population of *T. delbrueckii* may be due to strict anaerobic conditions and low invertase activity (Hanl, Sommer, & Arneborg, 2005; Visser, Scheffers, Batenburg-Van Vegte, & Dijken, 1990). However, the *T. delbrueckii* fermentation process is associated with aromatic metabolites, such as fruity esters, lactones, thiols, terpenes, and reduced amounts of unpleasant aromatic compounds, including higher alcohols (Taillandier, Lai, Julien-Ortiz, & Brandam, 2014).

3.2. Chemical composition

Microorganisms have essential roles in the fermentative process of coffee, namely mucilage removal and metabolite production. Specific nonvolatile and volatile compounds in coffee beans influence the quality of the beverage.

The nonvolatile compounds, including carbohydrates, organic acids, and alcohols, were monitored during coffee fermentation (Fig. 3). The low molecular weight carbohydrates are potential precursors of coffee beverage flavor and aroma. Their concentrations change depending on the type of processing used (Knopp, Bytof, & Selmar, 2006). The removal of coffee peel and mucilage during pulped coffee processing reduced the sucrose concentration (Koshiro, Jackson, Nagai, & Ashihara, 2015). Further, the hydrolysis of sucrose influences the predominance of glucose and fructose.

Acetic acid was produced throughout the fermentation process in all treatments. The presence of small amounts of this acid at the end of fermentation contributes to the acidity of the beverage (Evangelista et al., 2014a).

Citric, malic, and succinic acids are usually detected in green coffee (Evangelista et al., 2014). However, their concentrations were different in the region where the coffee was grown. Citric acid was detected in the fermentative process in region 2 in both processes. This acid is an intermediate compound in the plant's metabolic cycle, and when present at the end of fermentation, it may contribute to the acidity, fruity, and mild fruity flavor (Igamberdiev & Eprintsev, 2016).

Malic and succinic acids were detected at the beginning of fermentation. However, only succinic acid was observed until the end of the process in all treatments. Succinic acid is naturally present in coffee, and it contributes to the final acidity of the beverage (Buffo & Cardelli-Freire, 2004). The translocation of this acid occurs from the pulp to the grain with the aid of microbial activity, and it may cause changes in the sensory profile of the coffee beverage (Silva et al., 2013). Acidity

contributes to enhancing the sweetness of the coffee beverage (Farah, 2019). The reduction in malic acid concentration may have occurred through the conversion of malic acid to lactic acid by naturally occurring bacteria in coffee (Avallone, Brillouet, Guyot, Olguin, & Guiraud, 2002; Evangelista, Miguel, Silva, Pinheiro, & Schwan, 2015).

Lactic acid was detected in all treatments, and its presence improves the texture of the final beverage (Di Cagno, Minervini, Rizzello, De Angelis, & Gobbetti, 2011). The lactic acid concentration was inversely proportional to ethanol concentration. Only region 1 showed a high concentration of ethanol, and the other regions showed the dominance of lactic acid. The production of this acid seemed to have been affected by environmental factors, as the altitude that it had shown to affect produced ethanol faster through yeast (Silva, Schwan, Sousa Dias, & Wheals, 2000).

The pulped natural coffee inoculated with *T. delbrueckii* produced a higher amount of glycerol in the fermentation medium in region 1 (≈ 75). At the beginning of the fermentative process, high mono-saccharide concentrations could have stimulated the overproduction of glycerol by yeast (Wang, Zeng, Liu, & Yuan, 2006).

The profile of volatile compounds in coffee beans depends on the genetic variability, environmental condition, geographical origin, and pre- and post-harvest processing (Bertrand et al., 2012). The main volatile groups detected were amide, acids, alcohol, aldehyde, amine, ester, furan, hydrocarbon, ketone, lactone, phenol, pyran, pyrazine, pyridine, pyrrole, and thiophene.

Pyrrole and furan are volatile groups that differentiate the types of processing (Fig. 4). The removal of the coffee peel, pulp, and part of the mucilage stimulated the furan production in the pulped coffee process, and a higher pyrrole concentration was detected in the natural process. Pyrrole and furan are aromatic groups, with the former contributing a floral, rose-like, and smoky flavor and the latter contributing a caramel-like odor (Farah, 2019).

Specific volatile compounds (Complementary material) were produced in each region. The volatile compound 1-(3-methylpyrazine) ethanone from region 2 contributed a hazelnut flavor (Sunarharum, Williams, & Smyth, 2014). 3-Methylphenol and 2,3-dimethyl-2-cyclopenten-1-one from region 3 contributed nutty, sweet, and roasted flavors (Bressanello et al., 2017). Dihydro-3-methylene-2,5-furandione was detected in region 4. This volatile compound was detected in both processes, and it could be associated with coffee beans.

Specific compounds were produced by natural processing. 4-Methoxy benzenamine, 2,6-dimethyl-3-octylacetate, and 1,2-epoxy-3-propyl acetate were detected in region 1. Formic acid, 1-(2-hydroxy-5-methylphenyl) ethanone, and pantolactone were detected in region 2. These compounds contributed bitter, sweet, floral, and herbaceous flavors (Flament, 2001). In region 5, only methyl acetate was detected.

Among the volatiles detected, 2-furanmethanol was the most abundant compound detected in all treatments. Aside from contributing burnt, sweet, caramel, coffee, bitter, and astringent flavors, it also has antioxidant activity (Osada & Shibamoto, 2006). In natural coffee, both inoculations were detected to have benzyl alcohol, and 1-hydroxy-2-propanone. Inoculation with *T. delbrueckii* produced 2-phenyl-2-butenal, 5-amino-1-naphthol, 3,4-dimethyl-2-pentanone, and 1-hydroxy-2-propanone. *S. cerevisiae* inoculation produced 2-butanone.

Dodecanoic acid, hexadecenoic acid, benzyl alcohol, and 2-butanone were detected in pulped natural coffee inoculated with both yeasts. The volatile compounds produced during the fermentative process were 2-methyl-2-pentenal with *T. delbrueckii* and formic acid, 2-methyl-1,3-butadiene, 3,4-dimethyl-2-pentanone, and pantolactone with *S. cerevisiae*.

Benzyl alcohol, 2-phenyl-2-butenal, 2-pentenal, and 2-methyl aldehydes were detected in the inoculated treatments. These compounds contribute sweet, chocolate, fruity, and floral flavors, as well as the antioxidant activity of the final product (Yanagimoto, Ochi, Lee, & Shibamoto, 2004).

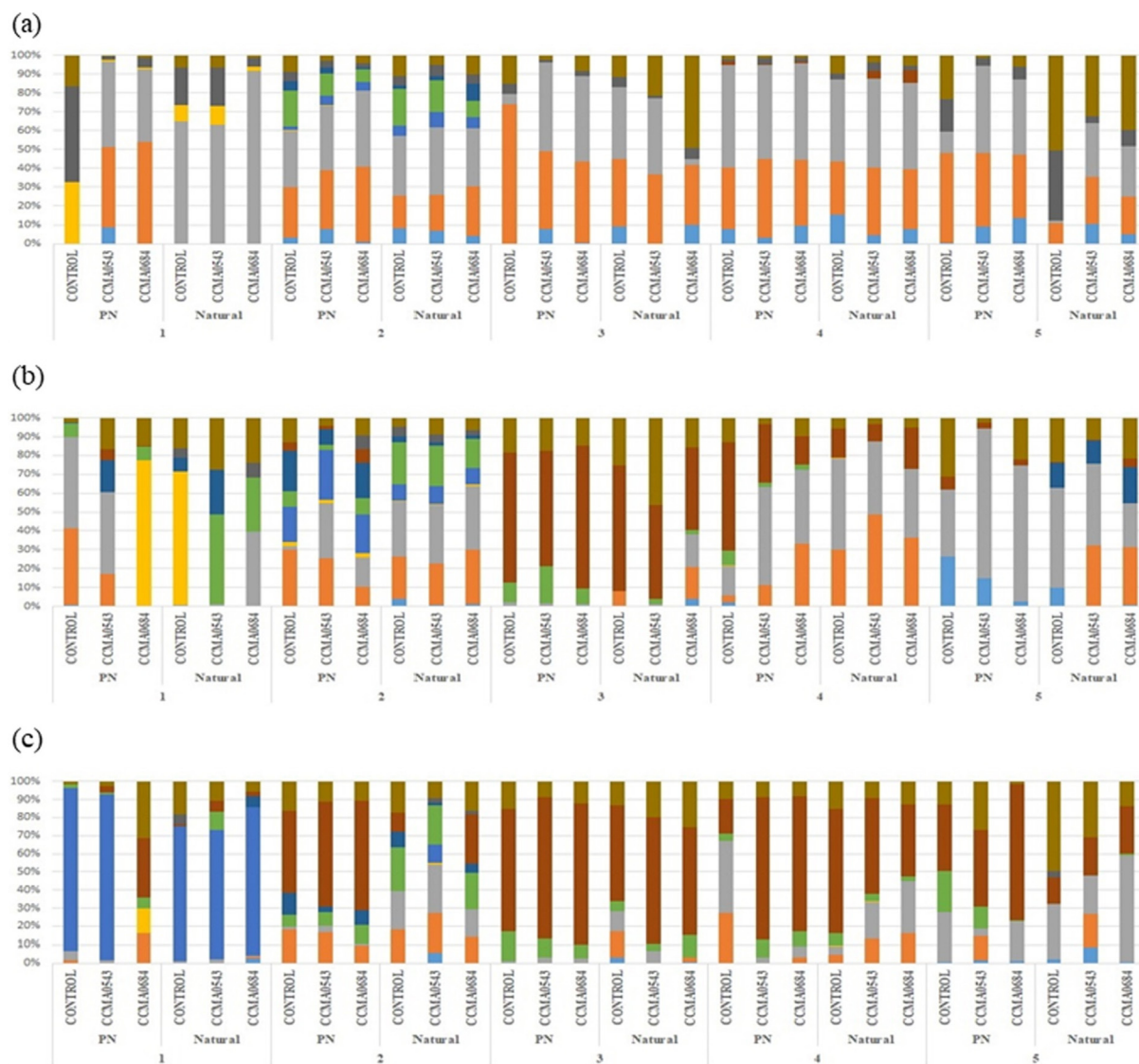


Fig. 3. Carbohydrates, organic acids, and ethanol were detected in the beginning (a), middle (b), and end (c) of coffee bean fermentation in region 1 (1), region 2 (2), region 3 (3), region 4 (4), and region 5 (5). ■ Sucrose, ■ Glucose, ■ Fructose, ■ Glycerol, ■ Ethanol, ■ Acetic acid, ■ Citric acid, ■ Lactic acid, ■ Malic acid, ■ Succinic acid. PN: pulped natural.

3.3. Sensory results

In addition to undergoing chemical analysis, the varieties of Bourbon amarelo, Catucaí amarelo, and Rubi cultivated at 1.161 m and 997 m, 885 m, and 470 m, and 997 m, respectively, were sensory evaluated in terms of body, acidity, sweetness, and finish. The preparation of the beverage was performed as proposed by the SCA, and the PCA evaluated the obtained data. The PC1 and PC2 components accounted for 89.19%, 88.39%, and 94.55% of not inoculated natural and pulped natural coffee, with *S. cerevisiae* (CCMA0543) and with *T. delbrueckii* (CCMA0684), respectively (Fig. 5).

The Bourbon amarelo and Catucaí amarelo cultivated in different regions showed different profiles of volatile compounds. The dynamic behavior of microorganisms changed the environmental conditions affected in the production of volatile compounds and the sensorial perceptions of the final beverage.

In spontaneous fermentation, the Bourbon amarelo cultivated in regions 1 and 3 showed better sensory scores in high altitudes in both processing. Catucaí amarelo cultivated in regions 2 and 5 showed the same sensory score for pulped coffee, but lesser altitude garnered a better sensory score for natural coffee. These coffees were characterized

by a lower phenol concentration and a higher pyran concentration. Low phenol concentration is desirable, as some compounds confer negative notes to coffee, such as brew, described as smoky, woody, phenolic, and meaty (Toci & Farah, 2014).

Yeast inoculation increased the coffee score compared with spontaneous fermentation. In spontaneous fermentation, natural coffee from regions 2 and 3 was classified as a commodity (79.25 and 79.75 points, respectively). Conversely, inoculation with *S. cerevisiae* and *T. delbrueckii* increased the grade to 83 and 85 points for region 2, respectively, and 82.25 points for region 3 for both inoculations.

Although both inoculations showed the same score in region 3, these coffees differed in the sensory attributes of the beverage. Natural coffee inoculated with *S. cerevisiae* were correlated with the body, and those inoculated with *T. delbrueckii* were correlated with finish, sweetness, and acidity.

T. delbrueckii had more influence on natural coffee, and the most significant variations in the score were obtained for regions 1 (87.25) and 2 (85) compared with those in spontaneous fermentation (83.25 and 79.25, respectively). Inoculation with *S. cerevisiae* mainly affected the pulped natural coffee and correlated the beverage with body, acidity, sweetness, and finish. The effectiveness of using the starter

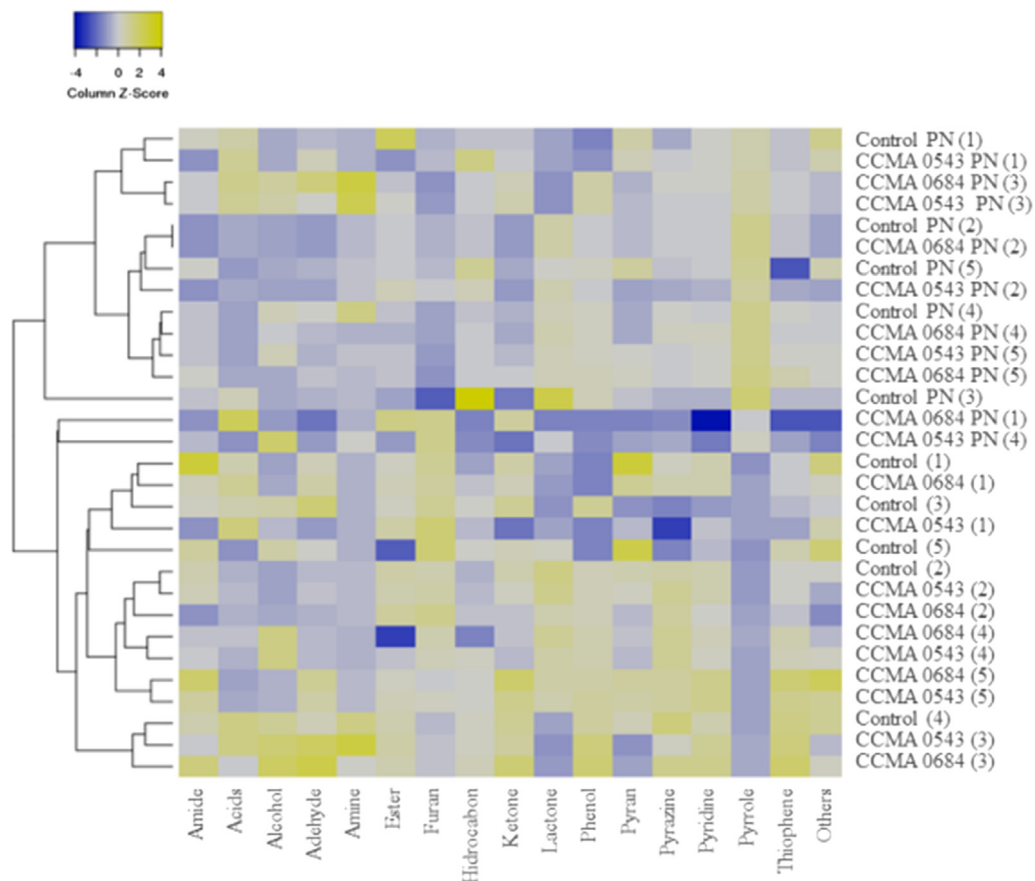


Fig. 4. Heatmap with normalized intensities of the volatile group showing a significant difference between spontaneous and inoculated fermentation in different regions. PN: pulped natural. Region 1 (1), Region 2 (2), Region 3 (3), Region 4 (4), and Region 5 (5).

culture technology positively improved the beverage quality (Evangelista et al., 2014a; Evangelista et al., 2014b; Martinez et al., 2017; Ribeiro et al., 2017a; Ribeiro et al., 2017b; Bressani et al., 2018; Martins, Ribeiro, da Miguel, Evangelista, & Schwan, 2019). The fermentation in bioreactor ensures the best performance of the starter cultures. Our results validated the methodology of batch anaerobic self-induced fermentation for coffee cherries from different varieties and fermentation conditions.

4. Conclusions

Environmental conditions affected the dynamical behavior of yeast

inoculation, the profiles of volatile and nonvolatile compounds, and the sensory perception of fermented coffee. Pyrrole and pyran can be used to evaluate the kind of processing the coffee beans underwent. Both yeasts increased the beverage score compared with the control. The coffee process and the type of inoculum to use should be evaluated in each region. Therefore, *S. cerevisiae* inoculation was the most suitable for pulped coffee, and *T. delbrueckii* inoculation was best used in natural coffee.

Declaration of Competing Interest

The authors declare that they have no known competing financial

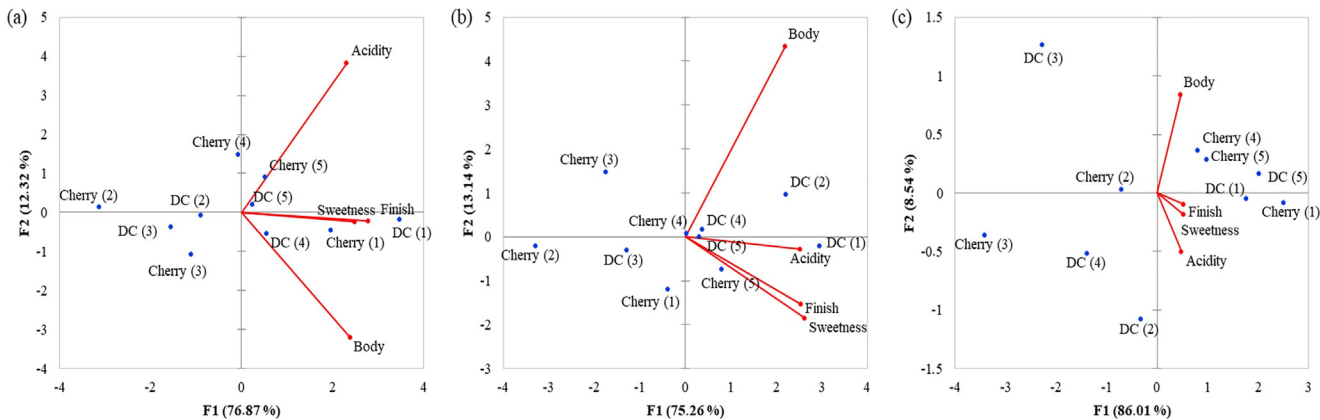


Fig. 5. PCA of the sensory attributes in natural coffee and pulped coffee by spontaneous fermentation (a) and inoculated with *S. cerevisiae* (CCMA0543) (b) and *T. delbrueckii* (CCMA0684) (c).

interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.foodres.2020.109482>.

References

- Avallone, S., Brillouet, J. M., Guyot, B., Olguin, E., & Guiraud, J. P. (2002). Involvement of pectolytic microorganisms in coffee fermentation. *International Journal of Food Science and Technology*, 37(2), 191–198. <https://doi.org/10.1046/j.1365-2621.2002.00556.x>.
- Batista, N. N., Ramos, C. L., Ribeiro, D. D., Pinheiro, A. C. M., & Schwan, R. F. (2015). Dynamic behavior of *Saccharomyces cerevisiae*, *Pichia kluyveri* and *Hanseniaspora uvarum* during spontaneous and inoculated cocoa fermentations and their effect on sensory characteristics of chocolate. *LWT - Food Science and Technology*, 63(1), 221–227. <https://doi.org/10.1016/j.lwt.2015.03.051>.
- Bertrand, B., Boulanger, R., Dussert, S., Ribeyre, F., Berthiot, L., Descroix, F., & Joët, T. (2012). Climatic factors directly impact the volatile organic compound fingerprint in green *Arabica* coffee bean as well as coffee beverage quality. *Food Chemistry*, 135(4), 2575–2583. <https://doi.org/10.1016/j.foodchem.2012.06.060>.
- Borrelli, I., Mataix, C., & Carrasco-Gallego, R. (2015). Smallholder farmers in the specialty coffee industry: Opportunities, constraints and the businesses that are making it possible. *IDS Bulletin*, 46(3), 29–44. <https://doi.org/10.1111/1759-5436.12142>.
- Brando, C. H. J., & Brando, M. F. (2014). Methods of coffee fermentation and drying. In R. F. Schwan, & G. H. Fleet (Eds.). *Cocoa and coffee fermentations* (pp. 367–396). Boca Raton, Florida, USA: CRC Press.
- Bressanello, D., Liberto, E., Cordero, C., Rubiolo, P., Pellegrino, G., Ruosi, M. R., & Bicch, C. (2017). Coffee aroma: Chemometric comparison of the chemical information provided by three different samplings combined with GC–MS to describe the sensory properties in cup. *Food Chemistry*, 214, 218–226. <https://doi.org/10.1016/j.foodchem.2016.07.088>.
- Bressani, A. P. P., Martinez, S. J., Evangelista, S. R., Dias, D. R., & Schwan, R. F. (2018). Characteristics of fermented coffee inoculated with yeast starter cultures using different inoculation methods. *LWT - Food Science and Technology*, 92(October 2017), 212–219. <https://doi.org/10.1016/j.lwt.2018.02.029>.
- Buffo, R. A., & Cardelli-Freire, C. (2004). Coffee flavour: An overview. *Flavour and Fragrance Journal*, 19(2), 99–104. <https://doi.org/10.1002/ffj.1325>.
- Cocolin, L., Bisson, L. F., & Mills, D. A. (2000). Direct profiling of the yeast dynamics in wine fermentations. *FEMS Microbiology Letters*, 189(1), 81–87. [https://doi.org/10.1016/S0378-1097\(00\)00257-3](https://doi.org/10.1016/S0378-1097(00)00257-3).
- CONAB. (2019). Campanha Nacional de Abastecimento. Retrieved September 13, 2019, from Campanha Nacional de Abastecimento website: [file:///C:/Users/mmmot/Downloads/CafZZAnliseZMensalZJaneiroZ2019_1\(1\).pdf](file:///C:/Users/mmmot/Downloads/CafZZAnliseZMensalZJaneiroZ2019_1(1).pdf). Acesso.
- Di Cagno, R., Minervini, G., Rizzello, C. G., De Angelis, M., & Gobbetti, M. (2011). Effect of lactic acid fermentation on antioxidant, texture, color and sensory properties of red and green smoothies. *Food Microbiology*, 28(5), 1062–1071. <https://doi.org/10.1016/j.fm.2011.02.011>.
- Díaz, C., Molina, A. M., Nähring, J., & Fischer, R. (2013). Characterization and dynamic behavior of wild yeast during spontaneous wine fermentation in steel tanks and amphorae. *BioMed Research International*, 2013, 1–13. <https://doi.org/10.1155/2013/540465>.
- Evangelista, S. R., Miguel, M. G. da C. P., Cordeiro, C. de S., Silva, C. F., Pinheiro, A. C. M., & Schwan, R. F. (2014a). Inoculation of starter cultures in a semi-dry coffee (*Coffea arabica*) fermentation process. *Food Microbiology*, 44, 87–95. <https://doi.org/10.1016/j.fm.2014.05.013>.
- Evangelista, S. R., Miguel, M. G. da C. P., Silva, C. F., Pinheiro, A. C. M., & Schwan, R. F. (2015). Microbiological diversity associated with the spontaneous wet method of coffee fermentation. *International Journal of Food Microbiology*, 210, 102–112. <https://doi.org/10.1016/j.jfoodmicro.2015.06.008>.
- Evangelista, S. R., Miguel, M. G. da C. P., Cordeiro, C. de S., Silva, C. F., Pinheiro, A. C. M., & Schwan, R. F. (2014b). Inoculation of starter cultures in a semi-dry coffee (*Coffea arabica*) fermentation process. *Food Microbiology*, 44, 87–95. <https://doi.org/10.1016/j.fm.2014.05.013>.
- Evangelista, S. R., Silva, C. F., Miguel, M. G. da C. P., Cordeiro, C. de S., Pinheiro, A. C. M., Duarte, W. F., & Schwan, R. F. (2014). Improvement of coffee beverage quality by using selected yeasts strains during the fermentation in dry process. *Food Research International*, 61, 183–195. <https://doi.org/10.1016/j.foodres.2013.11.033>.
- Farah, A. (2019). Coffee production, quality and chemistry (A. Farah, Ed.). Royal Society of Chemistry.
- Flament, I. (2001). Coffee flavor chemistry. Weinheim, Germany: Wiley-VCH.
- Hanl, L., Sommer, P., & Arneborg, N. (2005). The effect of decreasing oxygen feed rates on growth and metabolism of *Torulaspora delbrueckii*. *Applied Microbiology and Biotechnology*, 67(1), 113–118. <https://doi.org/10.1007/s00253-004-1695-2>.
- Igamberdiev, A. U., & Eprintsev, A. T. (2016). Organic acids: The pools of fixed carbon involved in redox regulation and energy balance in higher plants. *Frontiers Plant Science*, 7(2016JULY), 1–15. <https://doi.org/10.3389/fpls.2016.01042>.
- Jackels, S. C., & Jackels, C. F. (2005). Characterization of the coffee mucilage fermentation process using chemical indicators: A field study in Nicaragua. *Journal of Food Science*, 70(5), <https://doi.org/10.1111/j.1365-2621.2005.tb09960.x>.
- Knopp, S., Bytof, G., & Selmar, D. (2006). Influence of processing on the content of sugars in green *Arabica* coffee beans. *European Food Research and Technology*, 223(2), 195–201. <https://doi.org/10.1007/s00217-005-0172-1>.
- Koshiro, Y., Jackson, M. C., Nagai, C., & Ashihara, H. (2015). Sugar and organic acid content in *Coffea arabica* and *Coffea canephora* fruits Section Changes in the content of sugars and organic acids during ripening of *Coffea arabica* and *Coffea canephora* fruits. *Chem. Bull.*, 4(8), 378–383. <https://doi.org/10.17628/ECB.2015.4.378>.
- Lee, L. W., Cheong, M. W., Curran, P., Yu, B., & Liu, S. Q. (2015). Coffee fermentation and flavor - An intricate and delicate relationship. *Food Chemistry*, 185, 182–191. <https://doi.org/10.1016/j.foodchem.2015.03.124>.
- Legras, J. L., Galeote, V., Bigey, F., Camarasa, C., Marsit, S., Nidelet, T., ... Dequin, S. (2018). Adaptation of *S. cerevisiae* to fermented food environments reveals remarkable genome plasticity and the footprints of domestication. *Molecular Biology and Evolution*, 35(7), 1712–1727. <https://doi.org/10.1093/molbev/msy066>.
- Martinez, S. J., Bressani, A. P. P., Miguel, M. G. da C. P., Dias, D. R., & Schwan, R. F. (2017). Different inoculation methods for semi-dry processed coffee using yeasts as starter cultures. *Food Research International*, 102, 333–340. <https://doi.org/10.1016/j.foodres.2017.09.096>.
- Martins, P. M. M., Batista, N. N., Miguel, M. G. da C. P., Simão, J. B. P., Soares, J. R., & Schwan, R. F. (2020). Coffee growing altitude influences the microbiota, chemical compounds and the quality of fermented coffees. *Food Research International*, 129, <https://doi.org/10.1016/j.foodres.2019.108872>.
- Martins, P. M. M., Ribeiro, L. S., Miguel, M. G. da C. P., Evangelista, S. R., & Schwan, R. F. (2019). Production of coffee (*Coffea arabica*) inoculated with yeasts: impact on quality. *Journal of the Science of Food and Agriculture*, 99(13), 5638–5645. <https://doi.org/10.1002/jsfa.9820>.
- OIC. (2018). Segundo ano de excedentes previsto para 2018/19. Retrieved from Relatório sobre o mercado cafeeiro website: <http://www.ico.org/documents/cy2018-19/cmr-1218-p.pdf>.
- Osada, Y., & Shibamoto, T. (2006). Antioxidative activity of volatile extracts from Maillard model systems. *Food Chemistry*, 98(3), 522–528. <https://doi.org/10.1016/j.foodchem.2005.05.084>.
- Peñuela-Martínez, A. E., Carlos, Oliveros-Tascón, E., Juan, & Sanz-Urbe, R. (2010). Remoción del mucílago de café a través de fermentación natural. In *Cenicafé*, 61(2), 159–173.
- Ribeiro, L. S., Evangelista, S. R., Miguel, M. G. da C. P., van Mullem, J., Silva, C. F., & Schwan, R. F. (2018). Microbiological and chemical-sensory characteristics of three coffee varieties processed by wet fermentation. *Annals of Microbiology*, 68(10), 705–716. <https://doi.org/10.1007/s13213-018-1377-4>.
- Ribeiro, L. S., Miguel, M. G. da C. P., Evangelista, S. R., Martins, P. M. M., van Mullem, J., Belizario, M. H., & Schwan, R. F. (2017). Behavior of yeast inoculated during semi-dry coffee fermentation and the effect on chemical and sensory properties of the final beverage. *Food Research International*, 92, 26–32. <https://doi.org/10.1016/j.foodres.2016.12.011>.
- Ribeiro, L. S., Ribeiro, D. E., Evangelista, S. R., Miguel, M. G. da C. P., Pinheiro, A. C. M., Borém, F. M., & Schwan, R. F. (2017). Controlled fermentation of semi-dry coffee (*Coffea arabica*) using starter cultures: A sensory perspective. *LWT - Food Science and Technology*, 82, 32–38. <https://doi.org/10.1016/j.lwt.2017.04.008>.
- SCA. (2013). Cupping protocols. Retrieved May 21, 2019, from specialty coffee association of America. Website: <https://sca.coffee/research/protocols-best-practices>.
- Silva, Cristina F., Schwan, R. F., Sousa Dias, E., & Wheals, A. E. (2000). Microbial diversity during maturation and natural processing of coffee cherries of *Coffea arabica* in Brazil. *International Journal of Food Microbiology*, 60(2–3), 251–260. [https://doi.org/10.1016/S0168-1605\(00\)00315-9](https://doi.org/10.1016/S0168-1605(00)00315-9).
- Silva, C. F., Batista, L. R., Abreu, L. M., Dias, E. S., & Schwan, R. F. (2008). Succession of bacterial and fungal communities during natural coffee (*Coffea arabica*) fermentation. *Food Microbiology*, 25(8), 951–957. <https://doi.org/10.1016/j.fm.2008.07.003>.
- Silva, C. F., Vilela, D. M., Cordeiro, C. de S., Duarte, W. F., Dias, D. R., & Schwan, R. F. (2013). Evaluation of a potential starter culture for enhance quality of coffee fermentation. *World Journal of Microbiology and Biotechnology*, 29(2), 235–247. <https://doi.org/10.1007/s11274-012-1175-2>.
- Souza, M. L. de, Passamani, F. R. F., Ávila, C. L. da S., Batista, L. R., Schwan, R. F., & Silva, C. F. (2017). Uso de leveduras selvagens como agente de biocontrole contra fungos toxigênicos e produção de OTA. *Acta Scientiarum - Agronomy*, 39(3), 349–358. <https://doi.org/10.4025/actasciagron.v39i3.32659>.
- Sunarharum, W. B., Williams, D. J., & Smyth, H. E. (2014). Complexity of coffee flavor: A compositional and sensory perspective. *Food Research International*, 62, 315–325. <https://doi.org/10.1016/j.foodres.2014.02.030>.
- Taillandier, P., Lai, Q. P., Julien-Ortiz, A., & Brandam, C. (2014). Interactions between *Torulaspora delbrueckii* and *Saccharomyces cerevisiae* in wine fermentation: Influence of inoculation and nitrogen content. *World Journal of Microbiology and Biotechnology*, 30(7), 1959–1967. <https://doi.org/10.1007/s11274-014-1618-z>.
- Toci, A. T., & Farah, A. (2014). Volatile fingerprint of Brazilian defective coffee seeds: Corroboration of potential marker compounds and identification of new low quality indicators. *Food Chemistry*, 153, 298–314. <https://doi.org/10.1016/j.foodchem.2013.11.033>.

- 12.040.
- Vilela, D. M., Pereira, G. V. de M., Silva, C. F., Batista, L. R., & Schwan, R. F. (2010). Molecular ecology and polyphasic characterization of the microbiota associated with semi-dry processed coffee (*Coffea arabica* L.). *Food Microbiology*, 27(8), 1128–1135. [10.1016/j.fm.2010.07.024](https://doi.org/10.1016/j.fm.2010.07.024).
- Visintin, S., Ramos, L., Batista, N., Dolci, P., Schwan, F., & Cocolin, L. (2017). Impact of *Saccharomyces cerevisiae* and *Torulaspota delbrueckii* starter cultures on cocoa beans fermentation. *International Journal of Food Microbiology*, 257, 31–40. <https://doi.org/10.1016/j.ijfoodmicro.2017.06.004>.
- Visser, W., Scheffers, W. A., Batenburg-Van der Vegte, W. H., & Van Dijken, J. P. (1990). Oxygen requirements of yeasts. *Applied and Environmental Microbiology*, 56(12), 3785–3792.
- Wang, J. P., Zeng, A. W., Liu, Z., & Yuan, X. G. (2006). Kinetics of glucoamylase hydrolysis of corn starch. *Journal of Chemical Technology & Biotechnology: International Research in Process, Environmental & Clean Technology*, 81(4), 727–729. <https://doi.org/10.1002/jctb.1435>.
- Yanagimoto, K., Ochi, H., Lee, K. G., & Shibamoto, T. (2004). Antioxidative activities of fractions obtained from brewed coffee. *Journal of Agricultural and Food Chemistry*, 52(3), 592–596. <https://doi.org/10.1021/jf030317t>.